Identification of new allergens from olive (*Olea europaea*) pollen and mustard (*Sinapis alba*) seeds by using T7 Phage Display libraries printed on microarrays

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ABSTRACT

Type I allergy is a generalized health problem in industrialized countries, affecting about 30% of the population. Specifically, allergies to yellow mustard and olive pollen have a high clinical relevance in Mediterranean countries due to their high prevalence and the severity of their symptoms.
The specific IgE production is a key event in the sensitization process against allergens. This phenomenon can be exploited through a high-throughput screening technique to identify new allergens with diagnostic and therapeutic purposes.
Phage display technologies allow for the construction of phage libraries displaying cDNA from biological sources. We have constructed two T7 phage libraries, displaying the cDNA from yellow mustard seeds or olive pollen. Both libraries were iteratively biopanned four times in solution using mustard and olive-specific immobilized IgE to enrich them in IgE immunoreactive phages displaying allergens. After biopanning, 768 monoclonal phages of each library were printed on 20 nitrocellulose microarrays, and probed with sera from both sensitized patients. In total, we found 53 phages displaying allergenic peptides. Their immunoreactivity was analyzed by *dot blot.* The cDNA inserted in each phage was amplified by PCR and sequenced to identify unique phages. Then, we designed specific oligonucleotides to amplify the complete cDNAs of the most reactive allergens displayed on phages.
Our purpose is the characterization of new recombinant allergens from olive pollen and mustard seed in heterologous systems by cloning and sequencing their complete cDNAs. Recombinant variants, exhibiting the functional activities of the natural allergens, can be used for the development of more accurate molecular diagnosis and therapeutic approaches.

RESULTS

Identifying potential allergens by phage microarrays

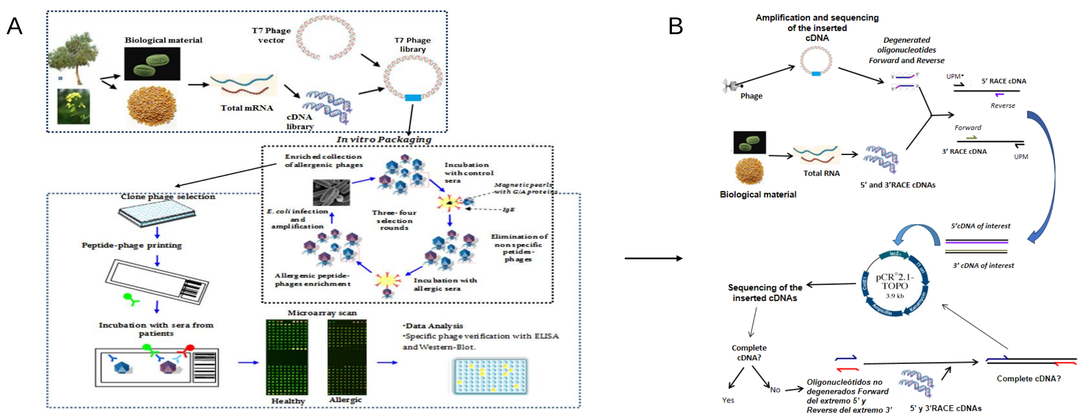


Figure 1. (A) Workflow to construct the two T7 phage libraries and to carry out the identification of those phages displaying allergenic peptides from mustard seeds and olive pollen. **(B)** Workflow to amplify the complete cDNAs of the most reactive allergens displayed on T7 phages.

Hybridization of phage microrrays with sera from mustard and olive pollen allergic patients

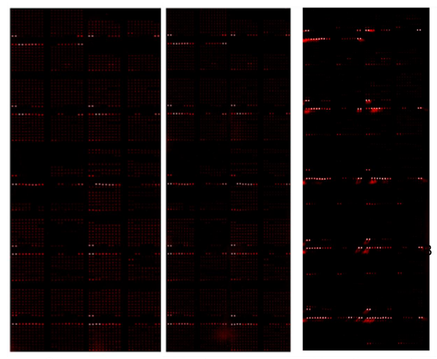
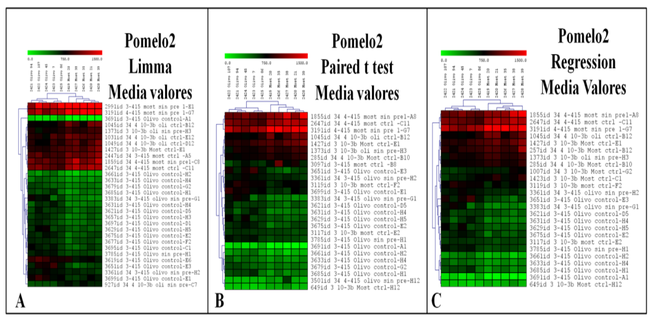


Figure 2. Images of the phage microarrays after hybridization with individual allergic patient's sera to mustard seeds or olive pollen. Anti-human IgE antibody and anti-mouse AlexaFluor647 were used as primary and secondary antibodies, respectively. The negative control corresponds to the incubation with the secondary antibody. Microarray scanning was made in the red cannel (647 nm)

Bioinformatic analysis of the microarrays



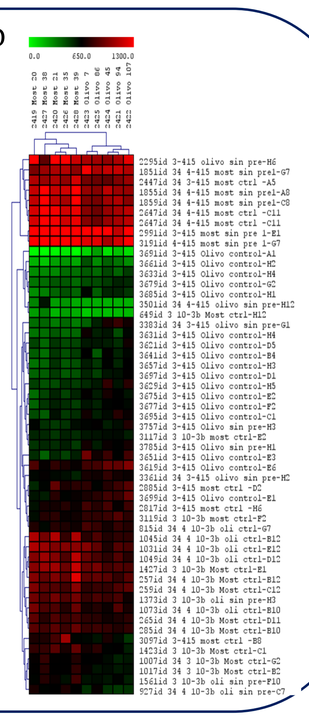


Figure 3. Statistical analysis was performed with several bioinformatic tools: *limma test* (A), *paired t-test* (B) and *regression test* (C) using Pomelo2 suite to identify the most reactive phages after normalization of the signal intensity. Supervised hierarchical cluster analysis of the most reactive phages was carried out with the MeV programme after statistical analysis (D). A total of 53 IgE immunoreactive phages were found in the screening, 31 displaying peptides from olive pollen and 22 from mustard seeds. **Red**, high reactivity. **Black**, moderate reactivity. **Green**, low reactivity.

Comparison of the signal intensity among the phages

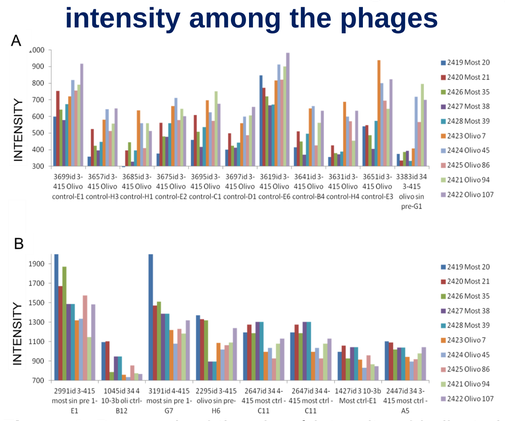


Figure 4. Bar graph of the signal intensity of indicated phages displaying immunoreactive peptides from olive pollen (A) and mustard seeds (B) after incubation with both types of allergic patient's sera. Phages displaying peptides from olive pollen showed stronger signal intensity than those displaying peptides from mustard seeds

Identification of phages displaying unique peptides by PCR amplification of the cDNA insert and sequencing

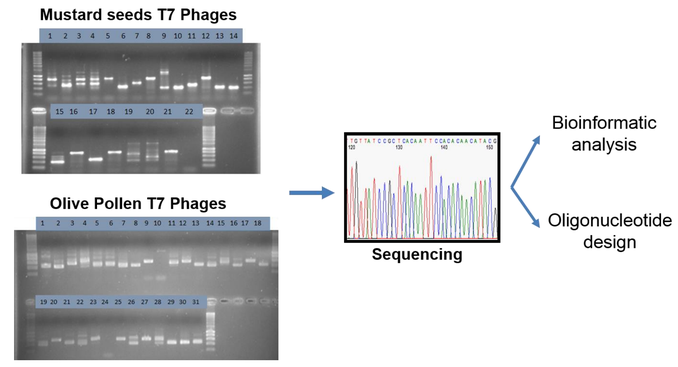


Figure 5. The cDNA insert of each reactive phage was amplified by PCR using the T7Up and T7Down oligonucleotides. Then, the amplified DNA was analysed in 1.5% agarose gels and the individual bands sequenced. This allowed for the identification of phages displaying unique amino acid sequences, bioinformatic analysis to determine the homology to proteins deposited in databases and the design of specific oligonucleotides to amplify the cDNA of the most immunoreactive proteins.

Verification of the immunoreactivity of unique phages

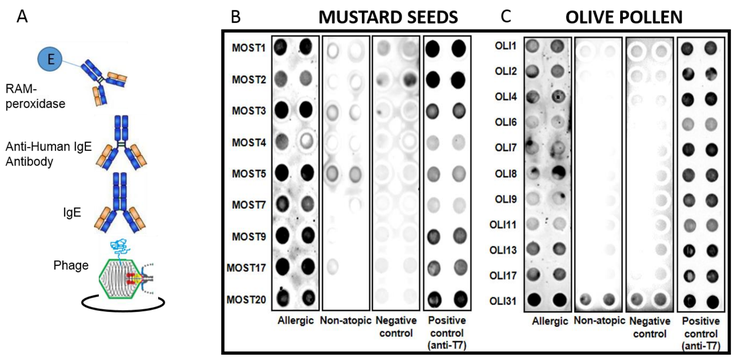


Figure 6. After phage amplification by *E. coli* infection and precipitation with polyethylenglycol, the immunoreactivity of the unique phages from mustard seeds (B) and olive pollen (C) was confirmed by *dot blot*. Each phage was tested in duplicate four times with i) rabbit polyclonal antiT7 antibody as positive control, ii) Only with the secondary antibody RAM-peroxidasas negative control, iii) indicated allergic patient's sera, and iv) non-atopic patient's sera as is depicted in (A).

PCR amplification, cloning and sequencing of the cDNAs encoding potential allergens

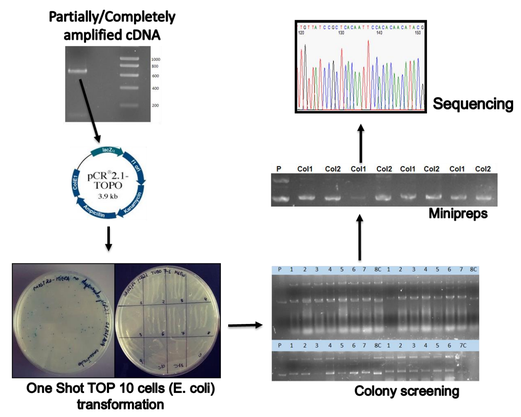


Figure 7. Workflow to clone and sequence PCR-amplified cDNAs using degenerate and non-degenerate oligonucleotides designed from the amino acid sequences displayed on phages. The cloning procedure was performed by using the TOPO®-TA Cloning® Kit (Invitrogen). TOP10 *E. coli* cells were transformed and individual colonies screened for plasmid preparation and sequencing with M13Fw or M13Rv oligonucleotides.

Amino acid sequences of the potential allergens

Table 1. Amino acid sequences of the potential allergens encoded by the cDNAs cloned and sequenced so far.

Sample name	Amino acid sequence displayed on the phage	Amino acid sequence encoded by the cDNA amplified*	NCBI Access Number, name and species
MOST5	GGKIRG	GGKIRGGEGHSSGDHKHDGEKKKKKDKKEKKHHDDGHHSSSSDSDSD-	BAJ08161.1stress protein (*Brassica napus* subsp. Chinensis)
MOST7	GKETNKDSLSNGERTGKSPA	QVRRDHPLSLSISISGGKETNKDSLSNGERTGKSPA-	XP_006405913.1 Hypothetical protein *(Eutrema salsugineum)*
MOST20	ITLKVVVFHLRRSSHLFYTSQVISQSRTRVKLNRVFFPR	MCRPSQTPHLTMSSARIDPPKRVLGLKEGVVTPPPIHGVSKITLKVVVFHLRRSSHLFYTSQVISQSRTRVKLNRVFFPR-	BAF01964.1 hypothetical protein *(Arabidopsis thaliana)*
MOST1	DSDLEAFSHNPAHGSFAPLAFQPSAMTNCANQRFLSY	VASSPDSDLEAFSHNPAHGSFAPLAFQPSAMTNCANQRFLSY-	XP_004162796.1 PREDICTED: uncharacterized protein LOC101232191 (*Cucumis sativus*)
OLI11	SAKKIK	SAKKIKKKFNIKGSDSSLDDCVYHLICPCCTLSQESRTLEMNNVQDGIWHGRGDTVCVGSYSEGNKALFEFSPPFAVSTESPECHTMQKTSINRLSTSIVVDSSPLVPSSD-	XP_002317257.2 hypothetical protein POPTR_0011s03990g *[Populus trichocarpa]*

* The amino acid sequence displayed on the phage is shown in red

CONCLUSIONS

- Statistical analysis of the normalized data obtained from the screened microarrays with sera from allergic patients to olive pollen and mustard seeds allowed us for the identification of 53 phages containing allergenic peptides. Most of these phages displayed cDNA from olive pollen.
- A total of 9 and 11 unique amino acid sequences were found in T7 phages displaying peptides from mustard seeds and olive pollen, respectively.
- The immunoreactivity of the unique amino acid sequences displayed on the T7 phages was confirmed by *dot blot* using sera from mustard and olive pollen allergic patients.
- One complete and three partial cDNA sequences encoding potentially allergenic proteins from *Sinapis alba* seeds have been cloned so far.
- Microarrays have been proved to be a useful tool for the identification of allergenic peptides or proteins displayed in T7 phages

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